

FLANGE AND JOINT INTEGRITY TOOLBOX TALK

A small joint error can become a major process leak.

PURPOSE AND SCOPE

Prevent process leaks caused by incorrect gaskets, misalignment, unsuitable bolts and incorrect tightening.

KEY HAZARDS

1  Incorrect gasket.	2  Damaged flange faces.	3  Misalignment.	4  Incorrect bolt grade.
5  Incorrect torque.	6  Uneven tightening.	7  Missing or damaged bolts.	8  Leakage under pressure.

CRITICAL POINTS TO VERIFY IN THE FIELD

01 Incorrect gasket.	02 Damaged flange faces.	03 Misalignment.
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REMEMBER — SERNEM

S

Stop and think

E

Evaluate the risk

R

Remove or reduce

N

Never assume

E

Ensure protection

M

Monitor and communicate



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FIELD REQUIREMENTS AND VERIFICATION

Prevent process leaks caused by incorrect gaskets, misalignment, unsuitable bolts and incorrect tightening.

SERNEM FIELD 01

FIELD VERIFICATION 1

Gasket correct

SERNEM FIELD 02

FIELD VERIFICATION 2

Flange faces sound

SERNEM FIELD 03

FIELD VERIFICATION 3

Alignment acceptable

SERNEM FIELD 04

FIELD VERIFICATION 4

Bolt grade correct

VERIFICATION BEFORE START

- | | |
|---|---|
| ☐ Gasket correct | ☐ Flange faces sound |
| ☐ Alignment acceptable | ☐ Bolt grade correct |
| ☐ Bolts sound | ☐ Torque tool calibrated |

SUPERVISOR DISCUSSION POINTS

- Is the gasket correct?
- Are flange faces sound?
- Is alignment acceptable?

FIELD APPLICATION



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CRITICAL CONTROL MEASURES

1. ELIMINATE / REDUCE THE RISK

Verify the correct gasket. • Use calibrated torque tools. • Perform final inspection before pressure.

2. ENGINEERING & PHYSICAL CONTROLS

Inspect flange faces. • Follow the approved tightening sequence. • Control first pressurisation with an exclusion zone.

3. WORK METHOD & FIELD CONTROLS

Confirm alignment before assembly. • Apply required lubrication conditions.

4. PPE, VERIFICATION & SUPERVISION

Use correct bolt and nut grades. • Record torque values.

CRITICAL WARNING



Team, flange integrity is more than tightening bolts. We will verify the gasket, flange faces, alignment, bolt grade and tightening sequence, and the system will not be pressurised until final inspection is complete.

SITE CHECKLIST — BEFORE STARTING WORK

- | | |
|--|--|
| ☐ Gasket correct | ☐ Flange faces sound |
| ☐ Alignment acceptable | ☐ Bolt grade correct |
| ☐ Bolts sound | ☐ Torque tool calibrated |
| ☐ Sequence correct | ☐ Torque record available |
| ☐ Final inspection complete | ☐ Crew briefed |



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ATTENDANCE AND APPROVAL

Project / Site: _____ Date: _____

Presented by: _____ Work area: _____

No	Full Name	Company / Role	Signature
1			
2			
3			
4			
5			
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9			
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12			